THE VIRTUAL MEET EXPERIENCE

2018 - 2019

HS VIRTUAL CHALLENGE MEET #1

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2018-2019 Virtual Challenge Meet #1 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.8.xx**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

Which of the following is equivalent to the expression `0xac + 0b1001001`?

A. 245_{10} B. 3301_4 C. 375_8 D. $e5_{16}$ E. All are equivalent

QUESTION 2

What is output by the code to the right?

A. `8` B. `20`
C. `33`
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
int j=8;
j*=3;
j+=9;
out.print(j);
```

QUESTION 3

What is output by the code to the right(e's indicate blank spaces)?

A. `212|0024|017` B. `212|24|17`
C. `024|0017|017` D. `017|0024|212`
E. `212|2400|170`

```
out.printf
  ("%2$03d|%1$04d|%0$03d",17,24,212);
```

QUESTION 4

What is output by the code to the right?

A. `is ultronA` B. `is ultron97`
C. `is ultron I` D. `all AI`
E. `is ultrona`

```
String s="all AI is ultron";
s+=s.charAt(4) + s.charAt(6);
s=s.substring(7);
out.println(s);
```

QUESTION 5

What is output by the code to the right?

A. `True` B. `False`
C. `false` D. `true`
E. There is no output due to a syntax error.

```
boolean b=true;
boolean a=false;
boolean c=!b^a || !a^b;
out.println(c);
```

QUESTION 6

What is output by the code to the right?

A. `6.0` B. `6`
C. `7.0` D. `7`
E. There is no output due to an error.

```
double g=6.5;
out.println(Math.floor(g));
```

QUESTION 7

What is the output by the code to the right?

A. `15` B. `420`
C. `-19` D. `15212`
E. `19`

```
int y="apples".compareTo("tacos");
if(y<0)
  out.print(15);
if(y>0)
  out.print(420);
else
  out.print(212);
```

QUESTION 8

What is the output by the code to the right?

A. `3.14` B. `3.1`
C. `3.142` D. `3.141`
E. `3`

```
System.out.printf("%.3f",Math.PI);
```

QUESTION 9

What is the sum of the ASCII values in `String s` to the right?

A. `1088` B. `1098`
C. `1066` D. `1112`
E. There is no output due to a syntax error.

```
String s="23weSDew43RRk ";
```

QUESTION 10

What is the output by the code to the right ?

A. `9` B. `0`
C. There is no output due to a syntax error.
D. There is no output due to a runtime error
E. Output cannot be determined until runtime.

```
int[]k=new int[8];
k[1]=4;
k[2]=6;
k[4]=7;
k[6]=9;
k[0]=k[k[k[1]]];
k[7]=k[k[2]];
k[3]=k[0]+k[7];
System.out.println(k[3]);
```

QUESTION 11

What is the output by the code to the right?

A. `nightS`
B. `Smb`
C. `September`
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
String c;
c="the 21st night of September";
Scanner s=new Scanner(c);
s.useDelimiter("\\W+|e");
for(int y=0;y<5;y++)s.next();
out.print(s.next());
s.next();
out.print(s.next());
```

QUESTION 12

What is the output by the code to the right ?

A. `30` B. `31`
C. `32` D. `27`
E. `0`

```
int y=128;
for(int x=0;x<8;x++) {
  y>>=1;
  y|=x*3;
  y++;
}
System.out.println(y);
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. `III, IV, II, I` B. `IV, III, II, I`
C. `II, IV, III, I` D. `I, II, III, IV`
E. `IV, III, II, I`

```
I. |(bitwise)
II. >>
III. ^{bitwise)
IV. &(bitwise)
```

QUESTION 14

What is the output by the code to the right ?

A. `4` B. `1`
C. `8` D. `2`
E. `6`

```
out.print(Long.SIZE/Short.SIZE);
```

QUESTION 15

What is the output by the code to the right ?

A. `[0, 420, 212, 15, 2, 9]`

B. `[0, 420, 212, 15, 2]`

C. `[9, 15, 17, 420, 2]`

D. `[9, 15, 17, 420, 2, 0]`

E. There is no output due to a syntax error.

```
ArrayList<Integer> a;
a = new ArrayList<Integer>();
a.add(9);a.add(15);a.add(212);
a.add(17);a.add(420);a.add(0);
Collections.reverse(a);
a.remove(2);
a.set(4,2);
out.println(a);
```

QUESTION 16

What is the output by the code to the right ?

A. `149` B. `150`

C. `213` D. `212`

E. There is no output due to a syntax error.

```
out.println(212|69&420^17);
```

QUESTION 17

What is the output by the code to the right ?

A. `F$L` B. `$FL`

C. `L$F` D. No output.

E. There is no output due to an error.

```
StringBuffer s=new StringBuffer();
s.append('F');
s.append('$');
s.append('L');
s.reverse();
out.println(s);
```

QUESTION 18

What could replace **`<1*>`** in the code to the right, so that the code will work as intended?

A. `<String,String>` B. `<>`

C. `<Object,Object>` D. `<Object,String>`

E. More than one of the choices above is correct.

QUESTION 19

What could replace **`<2*>`** in the code to the right, so that the code will work as intended?

A. `add` B. `put`

C. `push` D. `set`

E. There is no output due to a syntax error.

```
TreeMap<1*> tm;
tm = new TreeMap<1*>();
tm.<2*>("A", "apple");
tm.<2*>("B", "banana");
tm.<2*>("C", "crackers");
tm.<2*>("D", "domino");
tm.<2*>("E", "eggplant");
out.println(tm.get("D"));
```

QUESTION 20

What is the value of `0b1101010`?

A. 107_{10} B. 421_5 C. $6A_{16}$ D. $9A_{12}$ E. None of the above.

QUESTION 21

What is the output by the code to the right ?

A. `[212, 15, 17, 420]`

B. `[15, 420, 212, 17]`

C. `[420, 17, 15, 212]`

D. `[15, 212, 420, 17]`

E. There is no output due to a runtime error.

```
ArrayList<Integer>a;
a=new ArrayList<Integer>();
a.add(212);a.add(420);
a.add(17);a.add(15);
Collections.rotate(a, -3);
Collections.sort(a);
Collections.reverse(a);
Collections.rotate(a, 1);
out.println(a);
```

QUESTION 22

Which of the following could replace **<1*>** in the code to the right?

A. `abstract`
B. `super`
C. `public`
D. `interface`
E. `protected`

QUESTION 23

Which of the following could replace **<2*>** in the code to the right?

A. `abstract`
B. `super`
C. `public`
D. `interface`
E. `protected`

QUESTION 24

Assuming that **<1*>** and **<2*>** are filled correctly, what is the output by **//1** ?

A. `88`
B. `108`
C. `Television : $88.00`
D. `Television : $108.00`
E. This is no output due to an error.

QUESTION 25

Assuming that **<1*>** and **<2*>** are filled correctly, what is the output by **//2** ?

A. `false`
B. `true`
C. `Television : $88.00`
D. Output cannot be determined until runtime
E. This is no output due to a runtime error.

```
interface A{
  public double cost();
  public String product();
}
abstract class B implements A{
  int cost;
  double tax;
  String s;
  public B(int y, double t,String g){
     tax=t;
     cost=y;
     s=g;
  }
  <1*> double tax();
  public double cost() {
     return cost+cost*tax;
  }
  public String product() {
     return s;
  }
}
class C extends B {
  double discount;
  public C
  (double d,double t,int y,String g){
     super(y,t,g);
     discount=d;
  }
  public double cost() {
  return <2*>.cost()-<2*>.cost
                          *discount;
  }
  public String product() {
     return <2*>.product();
  }
  public double tax() {
     return <2*>.tax;
  }
}
/////////////client code/////////////
C c=new
       C(.2, .08, 100, "Television");
out.println(c.product()+" : "+
String.format("$%.2f", c.cost()));//1
C b=new C(.2, .08, 100,"Television");
out.println(c.equals(b));  //2
```

QUESTION 26

What is output by the code to the right?

A. `false true false`
B. `false true true`
C. `false false true`
D. `true true true`
E. `true false true`

```
out.print
("jebron_lam35".matches("\\w+")+" ");
out.print("GetGrinched".matches
                 ("[A-Z][a-z]{10}"));
out.println
  (" "+("420212".matches("......")));
```

QUESTION 27

Which of the following is a legal instantiation?

A. `HashSet h=new TreeSet();`

B. `List<Integer> j=new LinkedList<Integer>();`

C. `Object g=new String("abcd");`

D. `Map b=new TreeMap<Integer>();`

E. All of the above are illegal.

QUESTION 28

What is output by the line marked `//q28`?

A. `93`

B. `32`

C. `94`

D. `1`

E. `16`

QUESTION 29

What is output by the line marked `//q29`?

A. `0`

B. `2`

C. `-1`

D. `-2`

E. `1`

QUESTION 30

What is output by the line marked `//q30`?

A. `6142`

B. `1024`

C. `512`

D. `6123`

E. `1`

```
public static int recur(int h) {
  if(h>50)
    return 1;
  return 2*recur(h+6)+2;
}

//////////////////////////////////////
///////client code//////////////////
out.println(recur(25));  //q28
out.println(recur(55));  //q29
out.println(recur(-14));  //q30
```

QUESTION 31

What is the bit storage in the primitive type int?

A. `64` B. `8` C.`4` D. `16` E.`32`

QUESTION 32

Which of the lines to the right will first cause an error?

A. `//1`

B. `//2`

C. `//3`

D. `//4`

E. None of the above lines cause an error.

QUESTION 33

Assuming any error has been corrected, what is output by the code to the right ?

A. `[[2], 2]`

B. `[2, [2]]`

C. `[[2], moon]`

D. `[moon, [2]]`

E. There is no output due to a compile error.

```
Stack<String>s=new Stack(); //1
s.add(2+"");
s.add(s.toString()); //2
s.add("moon");
s.remove("2"); //3
Collections.sort(s); //4
out.println(s);
```

QUESTION 34

What could replace **<1*>** in the code to the right?

A. `v`
B. `val(t)`
C. `v%size`
D. More than one of the above.
E. None of the above.

QUESTION 35

What is outptut by the line marked **//1** in the client code to the right?

A. `636`
B. `54`
C. `635`
D. `55`
E. There is no output due to an error.

QUESTION 36

What is output by the line marked **//2** in the client code to the right ?

A. `212`
B. `cba`
C. `abc`
D. `ab1`
E. There is no output due to an error.

QUESTION 37

What is output by the line marked **//3** in the client code to the right ?

A. `ab1 cba 12 212 420 3E`
B. `abc cba 12 212 420 3E`
C. `abc ab1 cba 12 420 3E`
D. `abc ab1 12 212 420 3E`
E. There is no output due to a compile error.

QUESTION 38

What data structure is demonstrated by the code to the right ?

A. a linked list
B. a queue
C. a stack
D. a list
E. a hashtable

```
class Structure{
LinkedList<String>[] s;
int size;
public Structure(int t) {
  s = new LinkedList[t];
  size=t;
  for(int i=0;i<size;i++)
  s[i] = new LinkedList<String>();
}
public static int val(String s) {
  int y=0;
  for(char c:s.toCharArray()) {
  if(Character.isDigit(c))y+=c-'0';
  else
if(Character.isUpperCase(c))y+=c-'A';
  else
if(Character.isLowerCase(c))y+=c-'a';
  }
return y;
}
public void add(String t) {
  int v=val(t);
  s[<1*>].add(t);
}
public String remove(int pos) {
  return s[pos%size].poll();
}
public String toString() {
  String d="";
  for(int y=0;y<size;y++) {
  LinkedList<String> a=s[y];
          if(!a.isEmpty())
          for(int x=0;x<a.size();x++)
            d+=a.get(x)+" ";
  }
  return d.trim();
}
}
//////////////client code///////////
System.out.println
     (Structure.val("dog46CAT")); //1
Structure s=new Structure(12);
s.add("abc");s.add("ab1");
s.add("212");s.add("cba");
System.out.println(s.remove(15)); //2
s.add("3E");s.add("12");s.add("420");
System.out.println(s); //3
```

QUESTION 39

What is the runtime of deleting an item from a stack?

QUESTION 40

What is the pre-order traversal of the binary tree created when the following values are inserted in order ?

```
17, 212, 0, 420, 69, 8, -1
```

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2018-2019 Virtual Challenge Meet #1
Computer Science - KEY

1) A
2) C
3) C
4) B
5) C
6) A
7) D
8) C
9) B
10) A

11) B
12) A
13) C
14) A
15) B
16) C
17) C
18) E
19) B
20) C

21) B
22) A
23) B
24) C
25) A
26) E
27) C
28) C
29) E
30) A

31) E
32) E
33) C
34) C
35) B
36) C
37) A
38) E
39) O(1)
40) 17 0 -1 8 212 69 420

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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